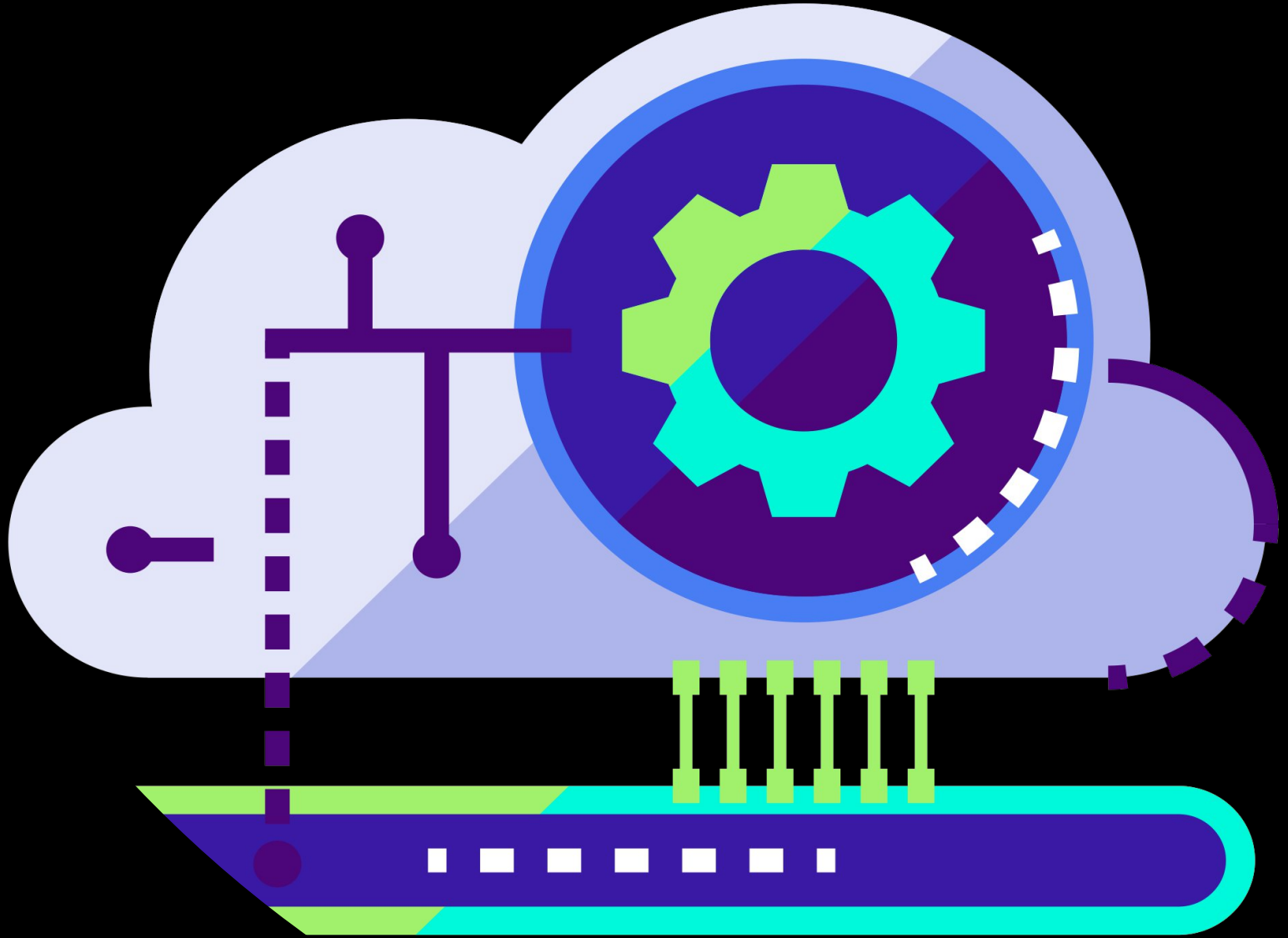


Central Agent Automation Engine (CAAE)

An adaptable agentic system which can
be extended to any problem/industry



Solving the Enterprise Agent Integration Bottleneck



Orchestration Logic

- Enterprise AI deployments typically suffer from integration problems where every new industry vertical or client tool forces a complete, costly rewrite of custom API adapters and state machines



Connection Layer

- Traditional agent implementations rely on brittle, hard-coded integrations to connect logic to external enterprise platforms.



Edge Execution Tier

- Data ingestion and operational modifications are tightly coupled with the core orchestration codebase.

Situation

Complication

- Open-ended LLM loops introduce severe operational risks, including unpredictable state routing, data hallucinations, and "AI slop".

- Maintaining custom connection tunnels across diverse cloud environments accelerates tech debt and degrades system performance

- Monolithic, industry-specific backends lack data agnosticism, making them impossible to rapidly scale across diverse use cases.

Resolution

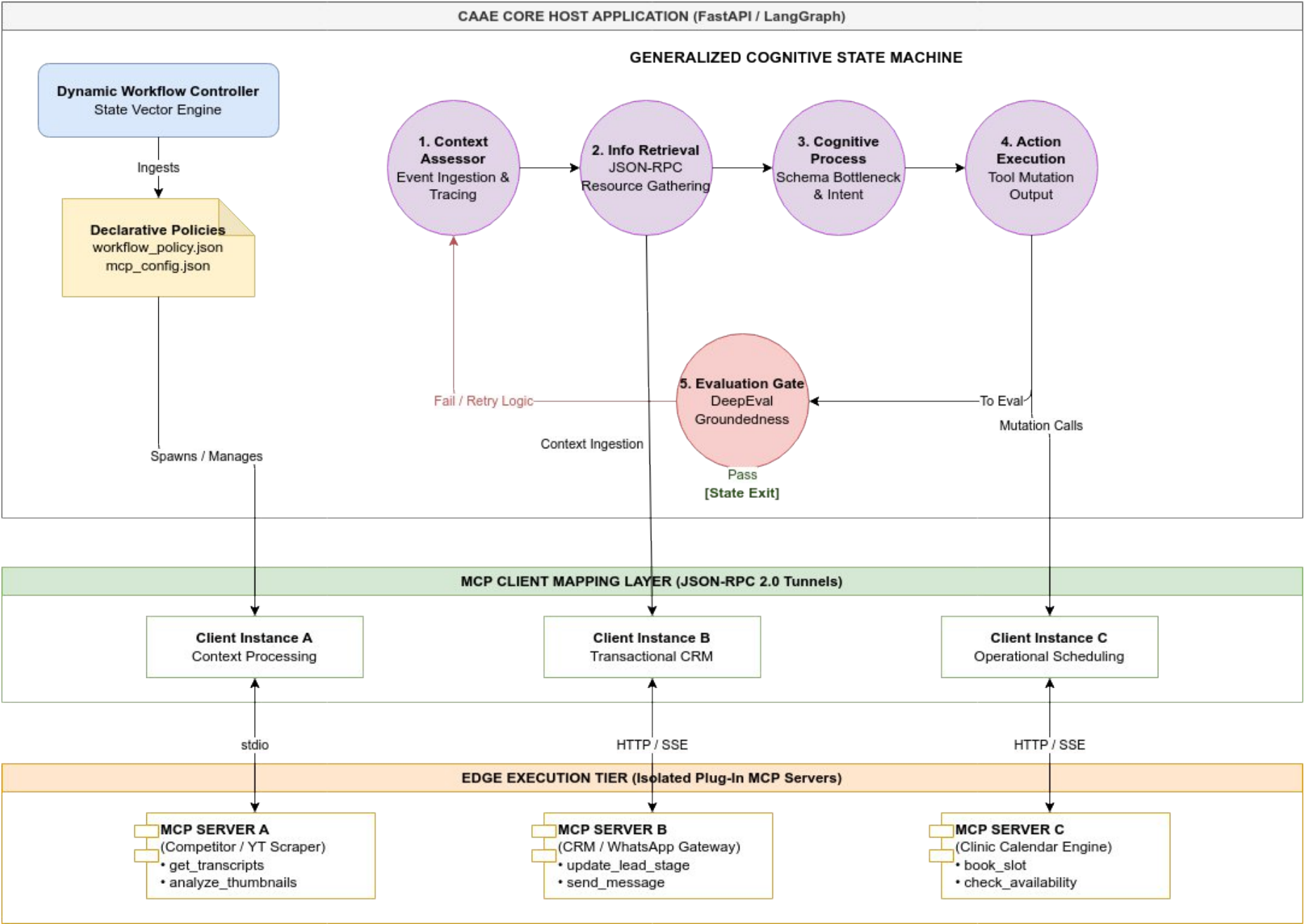
- Central Adaptable Automation Engine (CAAE) treats the core runtime purely as a stateful, data-agnostic host built on LangGraph.

- Establishes uniform, stateless MCP Server connections via JSON-RPC 2.0 capability negotiations and decouples communication mapping from execution via declarative `mcp_config.json` files.

- Deploys isolated, micro-service style plug-in servers that dynamically expose tools, resources, and schemas without visibility into the global graph.

Architecture Diagram

Modular, secure architecture with clear separation of roles and execution allowing for seamless extension and execution



Implementation roadmap

The project will be implemented in 4 distinct phases, with room for review and changes.

